Virtualisation

Ansible can be used to provision new machines and also configure them. Instead of using bare metal machines, you can create multiple virtual machines (VMs) on your system. Lots of free and open source software (FOSS) virtualisation software is available.

QEMU is a machine emulator and virtualiser. It can also use host CPU support to run guest VMs for better performance. It is written by Fabrice Bellard, and released under the GNU General Public License (GPL). You can install it on Parabola GNU/Linux-libre, using the following command:

```
$ sudo pacman -S qemu
```

KVM or kernel-based virtual machine has direct support in the Linux kernel. It requires hardware support to be able to run guest operating systems. It is written in C, and is released under the GNU General Public License.

You need to check if your hardware first supports KVM. The '*lscpu*' command will show an entry for '*Virtualization*' if there is hardware support. For example:

```
$ lscpu

Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                4
On-line CPU(s) list:   0-3
Thread(s) per core:    2
Core(s) per socket:    2
Socket(s):             1
NUMA node(s):          1
Vendor ID:             GenuineIntel
CPU family:            6
Model:                 78
Model name:            Intel(R) Core(TM) i5-6200U CPU @
2.30GHz
Stepping:              3
CPU MHz:               2275.341
CPU max MHz:           2800.0000
CPU min MHz:           400.0000
BogoMIPS:              4801.00
Virtualization:        VT-x
L1d cache:             32K
L1i cache:             32K
L2 cache:              256K
L3 cache:              3072K
NUMA node0 CPU(s):     0-3

Flags:                 fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr
sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtscp lm constant_
tsc art arch_perfmon pebs bts rep_good nopl xtopology
nonstop_tsc aperfmperf eagerfpu pni pclmulqdq dtes64 monitor
ds_cpl vmx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1
sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx
f16c rdrand lahf_lm abm 3dnowprefetch epb intel_pt tpr_shadow
vnmi flexpriority ept vpid fsgsbase tsc_adjust bmi1 avx2 smep
bmi2 erms invpcid mpx rdseed adx smap clflushopt xsaveopt
xsavec xgetbv1 xsaves dtherm ida arat pln pts hwp hwp_notify
hwp_act_window hwp_epp
```

You can also check the */proc/cpuinfo* output as shown below:

```
$ grep -E "(vmx|svm)" --color=always /proc/cpuinfo

flags           : fpu vme de pse tsc msr pae mce cx8 apic sep
mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2
ss ht tm pbe syscall nx pdpe1gb rdtscp lm constant_tsc art
arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc
aperfmperf eagerfpu pni pclmulqdq dtes64 monitor ds_cpl vmx
est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1 sse4_2
x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c
rdrand lahf_lm abm 3dnowprefetch epb intel_pt tpr_shadow vnmi
flexpriority ept vpid fsgsbase tsc_adjust bmi1 avx2 smep bmi2
erms invpcid mpx rdseed adx smap clflushopt xsaveopt xsavec
xgetbv1 xsaves dtherm ida arat pln pts hwp hwp_notify hwp_
act_window hwp_epp
```

The Libvirt project provides APIs to manage guest machines on KVM, QEMU and other virtualisation software. It is written in C, and is released under the GNU Lesser GPL. The virtual machine manager (VMM) provides a graphical user interface for managing the guest VMs and is written in Python.

You can install all this software on Parabola GNU/Linux-Libre using the following command:

```
$ sudo pacman -S libvirt virt-manager
```

A screenshot of VMM is provided in Figure 1.

Check your distribution documentation to install the appropriate virtualisation software packages.

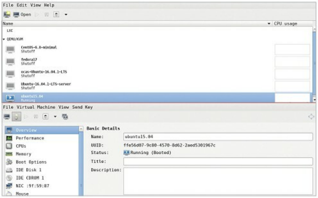

Figure 1: Virtual Machine Manager